PHASE I REVIEW

Real-Time Multimodal Driver Stress Monitoring and Adaptive Feedback via Biosignals and Conversational AI

Submitted By

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Mentor

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Problem Statement

Understanding a driver’s stress level is crucial for maintaining road safety, as elevated stress can impair judgment, slow reaction times, and increase the risk of accidents. Existing driver-monitoring systems often depend on visual cues or a single physiological signal, making them unreliable in real-world driving due to noise, motion artifacts, changing environments, and individual variability.

Stress, however, is a multidimensional physiological response involving cardiovascular, muscular, respiratory, and electrodermal systems. Relying on any single modality cannot fully capture this complexity.

This creates a need for a scientifically grounded, reliable, and continuous method to measure driver stress using multiple physiological inputs, especially during naturalistic driving where stress fluctuates dynamically.

Overview

This project aims to improve driver safety and well-being by providing a **reliable, real-time assessment** of stress levels using multimodal physiological data collected during naturalistic driving. Instead of depending on a single biosignal or camera-based cues, the system integrates **ECG, EMG, respiration, heart rate, and dual-channel GSR (hand + foot)** to capture a richer and more comprehensive representation of the driver's internal state.

By aligning and processing these diverse signals, the system can better handle real-world challenges such as motion artifacts, sensor noise, and inter-driver variability. This multimodal foundation enables a more accurate understanding of stress transitions across **low, medium, and high stress levels**.The system incorporates advanced preprocessing, stress-level labeling, classical machine learning baselines, and state-of-the-art deep learning architectures to capture both short-term temporal fluctuations and long-range physiological patterns.

In addition to building a robust stress detection backbone, the project lays the groundwork for **empathetic, context-aware driver support** through a conversational AI interface. Such feedback can enhance user trust, encourage mental relief, and form the basis of emotionally intelligent in-vehicle systems.

Beyond automobiles, the approach contributes to broader research in **physiology-driven affective computing**, with potential applications in healthcare, occupational safety, and next-generation human–machine interfaces. Ultimately, this work moves toward systems that not only monitor behaviour, but also interpret human emotional states in real time.

Literature Survey

Originating information		Framing of the research		Scope of the research			Research outcomes		
Paper	**Author/s**	**Purpose/Aims**	**Approach**	**Method of data collection**	**Study population/ sample size**	**Method and data collection limitations**	**Results**	**Discussion themes**	**Limitation of research outcomes**
Detecting Stress During Real-World Driving Tasks Using Physiological Sensors.	Jennifer A. Healey & Rosalind W. Picard (MIT Media Lab).	To determine if physiological sensors can detect driver stress levels during real-world driving conditions.	Multi-sensor data collection (ECG, EMG, respiration, skin conductance) on a set driving route with city, highway, and rest segments.	Real-time signals from 24 drives (each ~50 mins), annotated with video recordings and observer notes.	24 usable drives from multiple participants.	Not real-time inference or adaptive feedback; stress levels between defined segments were not well categorized.	97% classification accuracy for low/medium/high stress segments using 5-minute data intervals.	Future cars could modulate non-critical systems based on stress (e.g., dim screens, pause notifications).	Lacked adaptive interventions or real-time feedback; limited sensor deployment due to vehicle constraints.
An Explainable ML Approach Using Statistical Indexes and SVM for Stress Detection via EMG.	Olivia Vargas-Lopez, Carlos A. Perez-Ramirez, et al.	To evaluate the effectiveness of statistical time features (STFs) of EMG signals for stress detection using explainable SVM models.	Statistical feature extraction (mean, std, RMS, variance) from EMG + SVM classifiers with different kernels.	Experimental data collected from drivers using surface EMG sensors.	Sample size not clearly stated; controlled lab driving-like conditions assumed.	Single-modality; real-time applicability not tested; computational complexity concerns.	Achieved AUC of 0.97 with SVM (cubic kernel), indicating high prediction quality using few features.	Compared favorably to ECG/GSR-based systems; EMG is robust to skin moisture and heart conditions.	Less effective in dynamic, noisy real-world conditions; less holistic stress capture compared to multimodal.
Wearable and Automotive Systems for Affect Recognition from Physiology.	Jennifer A. Healey, advised by Rosalind Picard.	To create and evaluate wearable and car-mounted systems that recognize affective states, including stress, using physiological sensors.	Experimental designs across three settings (lab, ambulatory, automotive); used multimodal signals (ECG, EMG, GSR, respiration).	Over 36 hours of driving data; real-time sensors with annotated video + questionnaires.	10 drivers (including repeated drives); smaller sample for lab-based emotion study.	Affected by motion noise, individual variation, and limited sensor durability in ambulatory cases.	Stress detection rates of 96% (task-based validation), 89% (questionnaire-based), and strong correlation with video-coded events.	Early demonstration of affect-aware systems in vehicles; laid groundwork for emotion-sensitive interfaces.	Lack of real-time deployment; generalizability to wider populations untested.

Literature Survey

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Detection of Real-world Driving-Induced Affective State Using Physiological Signals and Multi-view Multi-task Machine Learning	Daniel Lopez-Martinez, Neska El-Haouij, Rosalind Picard.	To develop a multi-view, multi-task learning system that detects drivers' affective states (like stress) from physiological signals while accounting for individual drive variability and enabling interpretability.	Apply spectral clustering to create drive profiles, and use multi-task multiple kernel learning (MT-MKL) on EDA and heart rate data to classify high/low stress.	Used three real-world driving datasets (MIT drivedb, HciLab, AffectiveROAD) with physiological signals like EDA, ECG, HR. Stress was annotated via road type or subjective reports validated by drivers.	MIT drivedb: 17 drives, HciLab: 10 drives, AffectiveROAD: 13 drives	Only high and low stress levels (no medium) were classified. Limited sensor modalities (no facial/video input), and clustering labels relied partly on assumptions like city = high stress.	MT-MKL model improved performance compared to single-task models. Best accuracies: MIT drivedb = 93%, AffectiveROAD = 83%, HciLab = 71% (with RBF kernel, 3 clusters).	Importance of personalization in affect recognition. Need for emotionally intelligent driver interfaces. EDA as a key modality.	Did not explore intermediate (moderate) stress. Lacked multimodal inputs like facial cues or speech. Real-world deployment needs more real-time validation and richer contextual understanding.
Stress Evaluation in Simulated Autonomous and Manual Driving through the Analysis of Skin Potential Response and Electrocardiogram Signals	Pamela Zontone, Antonio Affanni, Riccardo Bernardini, Leonida Del Linz, Alessandro Piras, and Roberto Rinaldo.	To design and test a system that detects driver stress using electrodermal activity (SPR) and ECG signals in both autonomous and manual driving situations.	Subjects drove in a simulator through stress-inducing events under manual and autonomous conditions. Machine learning (SVM) was used to classify stress vs. non-stress states based on SPR features.	Two SPR signals (from both hands) and ECG were collected using wearable sensors. Motion artifacts were removed using a dual-channel MA filter. SPR features were extracted every 15 seconds and fed to an SVM.	13 healthy volunteers (8 males, 5 females), aged 19–35. Each participant completed both manual and autonomous drives with identical stress tasks.	Only SPR signals were used for ML classification due to variability in ECG linked to physical activity. Stress intervals were labeled based on task timing, not continuous perception.	Manual driving was perceived as more stressful, with more frequent SPR peaks. Autonomous driving showed fewer but higher SPR peaks during stress episodes. Average classification accuracy using SPR + SVM was **73%**.	Manual driving results in more frequent physiological stress responses, while autonomous driving leads to fewer but more intense reactions. SPR-based classifiers can help assess driver acceptance of automation.	Used fixed labels for stress periods based on task timing, which may not reflect actual emotional fluctuations. The sample size was relatively small, and ECG data was not included in the classification model.
Detecting Driving Stress in Physiological Signals Based on Multimodal Feature Analysis and Kernel Classifiers	L.-L. Chen, Y. Zhao, P.-F. Ye, J. Zhang, and J.-Z. Zou.	To design a driving stress detection system using multimodal physiological features and kernel-based classifiers, aiming to identify different levels of stress and support adaptive in-vehicle stress management systems.	Used ECG, GSR, and RESP signals; extracted features from time, frequency, and wavelet domains; employed Sparse Bayesian Learning and PCA for dimensionality reduction; implemented kernel classifiers in both offline and continuous stress evaluation modes.	Collected from 14 drivers during real-world driving on a fixed route including rest, highway, and city segments. Physiological signals (ECG, GSR, RESP) were continuously recorded and segmented for analysis.	14 drivers participated in real driving experiments under varying road conditions (rest, highway, city).	Small sample size (14 drivers); variations in individual physiological responses reduced generalization. Driving conditions were predefined, which may limit model robustness in unpredictable traffic.	Achieved over 99% accuracy in drive-level classification and 89% accuracy in cross-drive validation using 10-second signal windows. The system reliably tracked stress changes aligned with traffic transitions.	Highlights the value of multi-view data fusion and hierarchical clustering for personalizing stress models. Suggests the future of real-time emotion-aware systems embedded in vehicles for improved user safety and comfort.	Limited dataset restricts scalability; models may not generalize across diverse populations or driving scenarios. Continuous deployment in real-time systems remains a challenge without further development.

Proposed Idea

Limitations of Existing Systems

- Rely mainly on **single signals** like ECG or facial expression, ignoring whole-body stress.
- Fail in **real-world driving** due to motion, sensor noise, and environmental interference.
- Give **no helpful response** to the driver; just classify stress without action.
- Are **static and generic**, not personalized for individual drivers or preferences.
- Can be **distracting or irritating**, especially with poorly timed or robotic alert

Our Multimodal Feedback-Based Solution

- Uses **multiple signals** (ECG, EMG, RESP, GSR, HR) for a richer, more reliable view of stress.
- Employs **advanced sequence models** (CNN-BiLSTM-Attention, CNN-GRU-Attention, etc.) for robust temporal stress recognition.
- Provides **context-aware feedback** through a lightweight, gentle LLM interface. Helps in real-time with **breathing cues, calming music, or optional detours**.
- Learns from the driver to offer **personalized, adaptive, and non-intrusive support** over time.
- Generates **stress trend visualizations and heatmaps**, contributing to safer routes and supporting broader intelligent transportation insights.

Proposed Work

Multimodal Stress Monitoring Framework	Develop a unified system that continuously captures and processes **six physiological modalities**, ECG, EMG, Respiration, Heart Rate, Hand-GSR, and Foot-GSR, to enable a richer, more reliable understanding of driver stress.
Deep Learning - Driven Stress Prediction	Design, train, and evaluate advanced **sequence-learning architectures** to improve classification accuracy, robustness to noise, and temporal sensitivity compared to traditional ML baselines.
Real-Time Inference Pipeline	Integrate the best-performing model into a **real-time processing pipeline** capable of handling live, continuous multimodal data streams, producing low-latency stress predictions suitable for in-vehicle deployment.
Conversational Feedback Interface (LLM-Assisted)	Implement an early-stage **lightweight conversational module**, enabling empathetic, non-intrusive, context-aware feedback tailored to the driver's current stress level, paving the way for future adaptive interactions.
Adaptive Control & Feedback Loop	Develop a feedback control layer that dynamically links predicted stress levels with the conversational system, allowing **graduated responses** (e.g., supportive prompts, reminders, brief guidance) based on stress intensity, trend, or sudden spikes.

Methodology

1. Data Acquisition

- **Dataset:** Stress Recognition in Automobile Drivers (PhysioNet)
- **Participants:** 17 drivers, ~22 hours of naturalistic driving
- **Modalities:** ECG, EMG, Respiration, HR, GSR (hand & foot)
- **Sampling Rates:**
  - ECG ~31 Hz
  - EMG ~15 Hz
  - GSR(hand/foot) ~4 Hz
- **Driving Stress Levels:** Baseline (Low), Moderate, High
- **Raw → Resampled:** All modalities aligned and unified for later ML/DL pipelines

2. Preprocessing

Filtering:

- **ECG:** 0.5–40 Hz band-pass
- **EMG:** Full-wave rectification + RMS smoothing
- **Respiration:** 0.1–0.5 Hz
- **GSR:** Low-pass ≤1 Hz

Methodology

Resampling: All signals resampled to 100 Hz for cross-modality sync

Segmentation:

- 5-min non-overlapping windows (Classical ML)
- 30-sec windows, 50% overlap (Deep Learning)

Noise Handling: Percentile outlier clipping + median imputation

Normalization:

- **ML:** Robust Scaling
- **DL:** Per-channel Z-score normalization

3. Feature Extraction (Classical ML Path)

- **ECG/HRV:** Mean HR, SDNN, RMSSD, pNN50, LF/HF ratio, RR variability, SD1–SD2
- **EMG:** Mean amplitude, RMS, skewness/kurtosis, zero-crossings
- **Respiration:** Breathing rate, variability, spectral energy, peak count
- **GSR (Hand/Foot):** Mean conductance, std, peak count, slope
- **Multimodal Feature Vector:** ~29 engineered features per 5-min window
- **Storage:** Session-level structured CSVs for reproducibility

Methodology

4. Stress Labeling

- **Low Stress:** Baseline driving
- **Medium Stress:** Moderate traffic & navigation load
- **High Stress:** Dense traffic, complex maneuvers
- **Label Assignment:**
  - ML windows inherit scenario label directly
  - DL windows use **majority voting** within each 30-sec window

5. Deep Learning Input Construction

- **Tensor Representation:**

  $X \in \mathbb{R}^{T \times C}$ with

  - **T = 3000 samples** (30 s × 100 Hz)
  - **C = 6 channels** (ECG, EMG, RESP, HR, GSR-H, GSR-F)
- **Purpose:** Enables sequence-learning and cross-modal temporal interaction
- **Models Supported:** CNN-BiLSTM-Attention, CNN-GRU-Attention, ResNet1D-LSTM, Inception1D-LSTM, Parallel CNN-LSTM

Methodology

5. Model Development & Classification

A. Classical Machine Learning Pipelines

1. Ensemble ML (Soft Voting Classifier)

- Base learners: **Random Forest + Gradient Boosting + RBF-SVM**
- Probabilistic soft voting for stable decisions

2. Stacking Ensemble (Final Best Classical Model)

- Level-0 Learners: **LightGBM + CatBoost** (fallback: Random Forest)
- Level-1 Meta-Learner: **Logistic Regression**
- Pipeline:
    - Median imputation
    - Percentile clipping
    - Robust scaling
    - **SMOTE** for class balance
    - **5-fold cross-validation**
- **Outcome:** Strong, interpretable baseline; limited temporal modeling ability

Methodology

B. Deep Sequence Models

- **Architectures Implemented:**
  - CNN-BiLSTM-Attention
  - CNN-GRU-Attention (lightweight + high accuracy)
  - ResNet1D-LSTM
  - Inception1D-LSTM
  - Parallel CNN-LSTM

- **Training Setup:**
  - Mixed-precision FP16
  - Adaptive LR scheduling
  - Early stopping
  - GPU-accelerated (RTX 4060)
- **Outcome:** Captures temporal patterns missed by ML baselines

Methodology

7. Feature & Tensor Storage

- **Feature Matrices:** Classical ML (CSV format)
- **Deep Tensors:** Saved as structured numpy arrays preserving temporal order
- **Aggregated Datasets:** For cross-driver and global evaluations
- **Ensures:** Traceability, consistency, and reproducible experimentation

8. Future Enhancements

- **Advanced Deep Models:** Transformers for long-range temporal modeling
- **Fuzzy Logic Layer:** For adaptive stress scoring under uncertain conditions
- **Conversational Feedback:**
  - Lightweight, safety-aware LLM
  - Delivers calming prompts, breathing cues, and adaptive messages
  - Integrates with real-time stress prediction pipeline

Dataset Description

Dataset Summary:

- There is a total of 17 different driver recordings.
- Each recording is of roughly 1 hour to 1 hour 20 minutes.
- The total recorded time is 22 hours 14 minutes.
- Stress Recognition in Automobile Drivers.

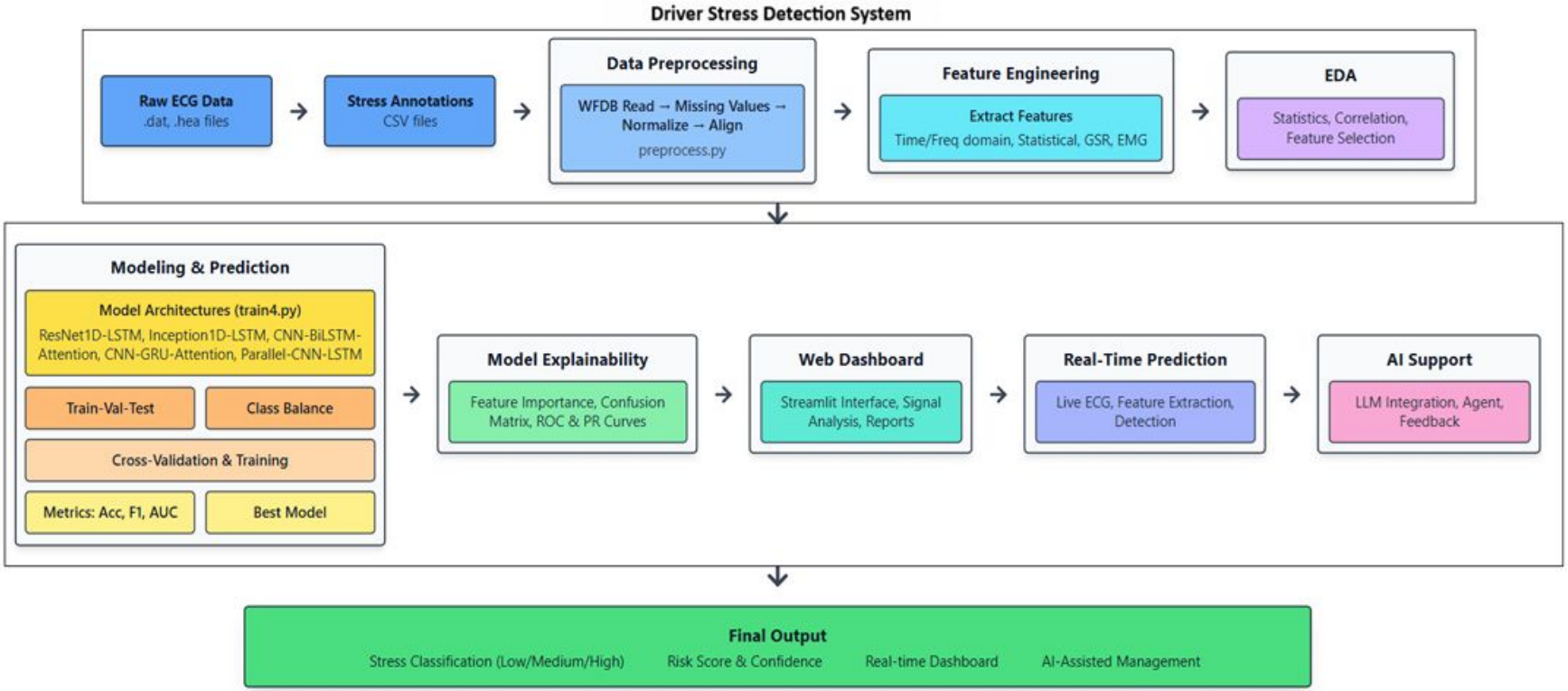
Feature information:

ECG, EMG, Foot GSR, Hand GSR, Heart Rate, Respiration

Sampling frequency :

15 Hz, 15.5 Hz and 31 Hz used

Block Diagram



Scope and Feasibility

- The project focuses on developing a real-time stress prediction system using multimodal physiological signals, including ECG, EMG, respiration, heart rate, and hand/foot GSR from wearable sensors.
- It involves the design and training of a deep learning model for stress classification using existing dataset and integration of a lightweight LLM-based conversational module to provide adaptive feedback based on predicted stress levels.
- The system will be implemented as a prototype with a web application/application showcasing the signal readings and the stress analysis along with LLM responses.
- The system has potential to be fully integrated into the smart system of a vehicle to be utilized as an AI assistant personalized for the driver that monitors the driver and helps manage stress levels.
- It can also be used to map out a stress heatmap of the roads by taking stress data from multiple vehicles which can be utilized for infrastructure planning to find out high stress road areas and intersections.
- This stress heatmap can also be integrated into maps to allow users to navigate a path of least stress as an additional option.

Demo of implementation

```
start,end,label
0.0,300.0,low
300.0,600.0,low
600.0,900.0,low
900.0,1200.0,low
1200.0,1500.0,low
1500.0,1800.0,low
1800.0,2100.0,low
2100.0,2400.0,low
2400.0,2700.0,low
2700.0,3000.0,low
3000.0,3300.0,low
3300.0,3600.0,low
3600.0,3900.0,low
```

Label file

Start: beginning of segment

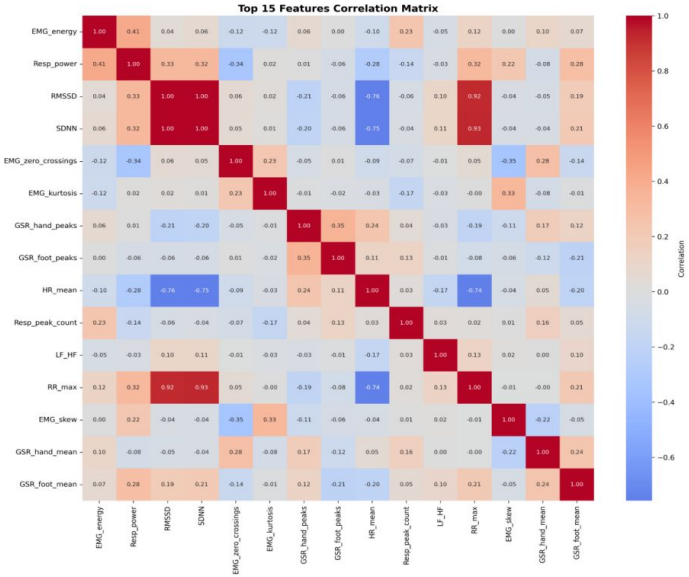
End: end of segment

Label: stress level

Demo of implementation

```
HR_mean,SDNN,RMSSD,pNN50,LF_HF,RR_mean,
RR_std,RR_min,RR_max,EMG_mean,EMG_std,
EMG_energy,EMG_skew,EMG_kurtosis,EMG_rms,
EMG_zero_crossings,GSR_foot_mean,
GSR_foot_std,GSR_foot_peaks,GSR_foot_slope,
GSR_hand_mean,GSR_hand_std,
GSR_hand_peaks,GSR_hand_slope,Resp_rate,
Resp_std,Resp_power,Resp_peak_count,label
```

Feature extraction and feature engineering has been performed.



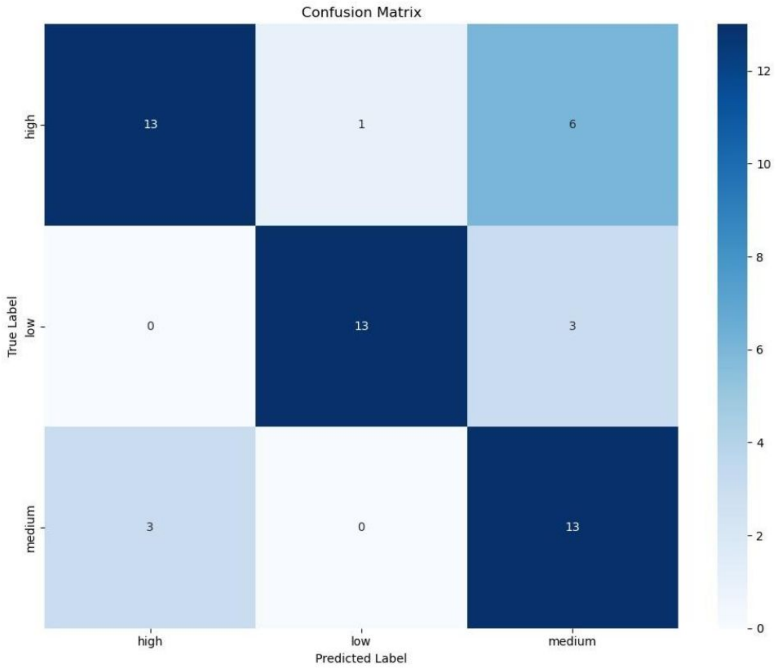
Feature Correlation matrix was generated.

Classical Machine Learning Models

Model Evaluation

Ensemble with Randomforest, GradientBoostingClassifier, SVC using soft voting

Confusion Matrix



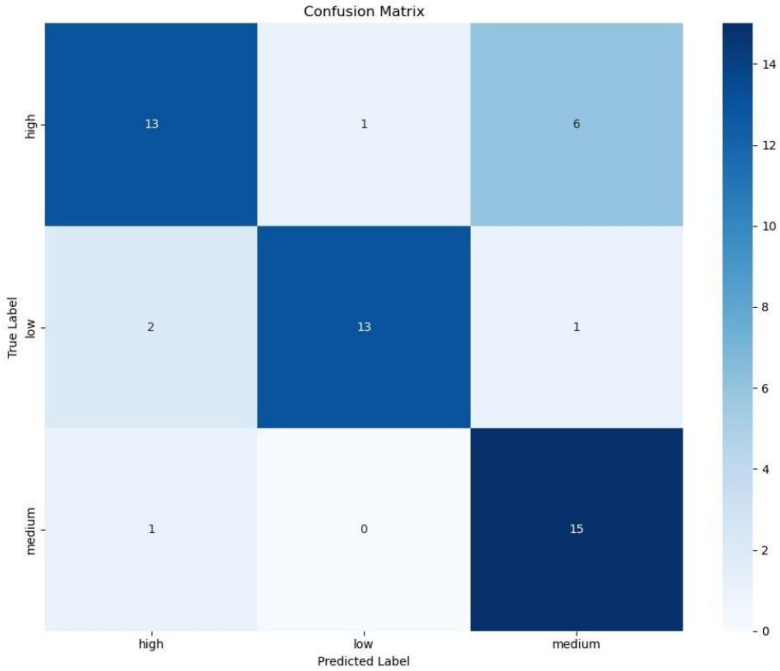
Classification Report

	precision	recall	f1-score	support
high	0.81	0.65	0.72	20
low	0.93	0.81	0.87	16
medium	0.59	0.81	0.68	16
accuracy			0.75	52
macro avg	0.78	0.76	0.76	52
weighted avg	0.78	0.75	0.75	52

Model Evaluation

Stacking Ensemble with LightGBM and CatBoost with Logistic Regression as Meta Learner

Confusion Matrix



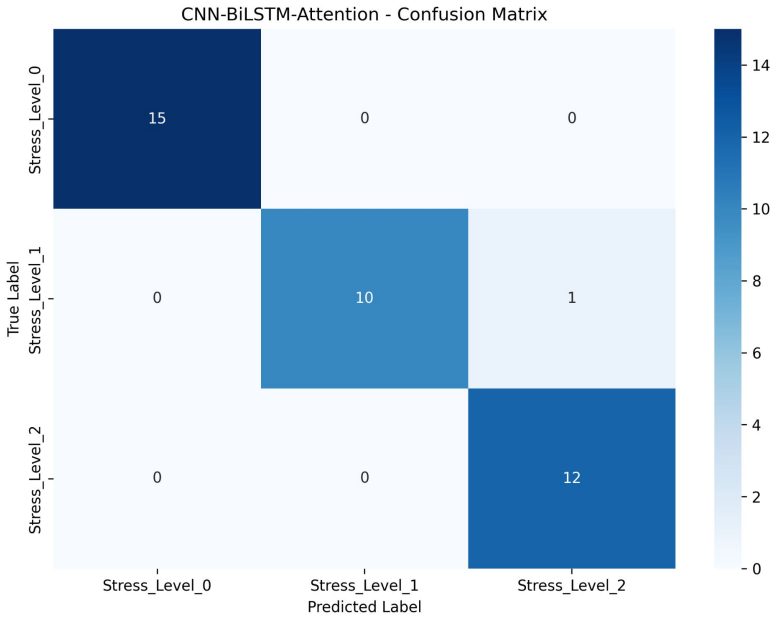
Classification Report

	precision	recall	f1-score	support
high	0.81	0.65	0.72	20
low	0.93	0.81	0.87	16
medium	0.68	0.94	0.79	16
accuracy			0.79	52
macro avg	0.81	0.80	0.79	52
weighted avg	0.81	0.79	0.79	52

Deep Learning Models

Model Evaluation

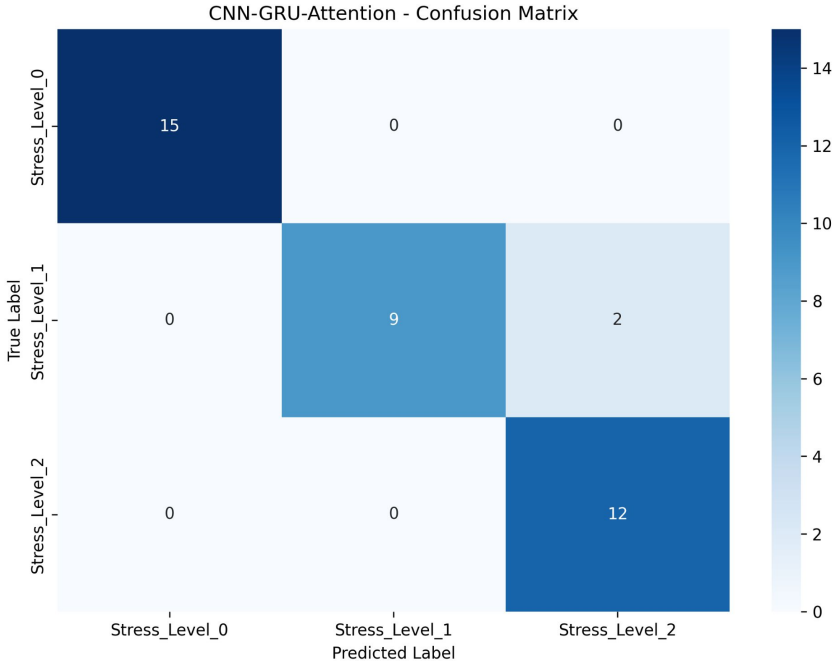
CNN - BiLSTM - Attention



	precision	recall	f1-score	support
Stress_Level_0	1.00	1.00	1.00	15
Stress_Level_1	1.00	0.91	0.95	11
Stress_Level_2	0.92	1.00	0.96	12
accuracy			0.97	38
macro avg	0.97	0.97	0.97	38
weighted avg	0.98	0.97	0.97	38

Model Evaluation

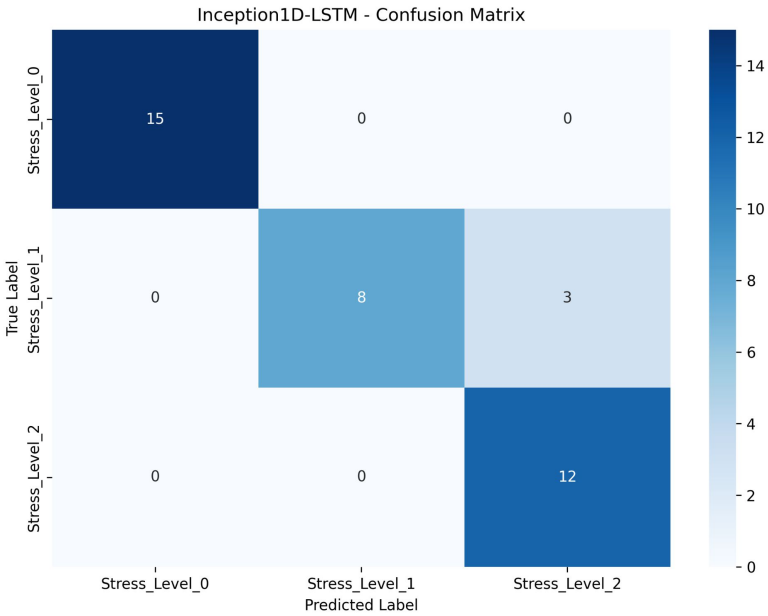
CNN - GRU- Attention



	precision	recall	f1-score	support
Stress_Level_0	1.00	1.00	1.00	15
Stress_Level_1	1.00	0.82	0.90	11
Stress_Level_2	0.86	1.00	0.92	12
accuracy			0.95	38
macro avg	0.95	0.94	0.94	38
weighted avg	0.95	0.95	0.95	38

Model Evaluation

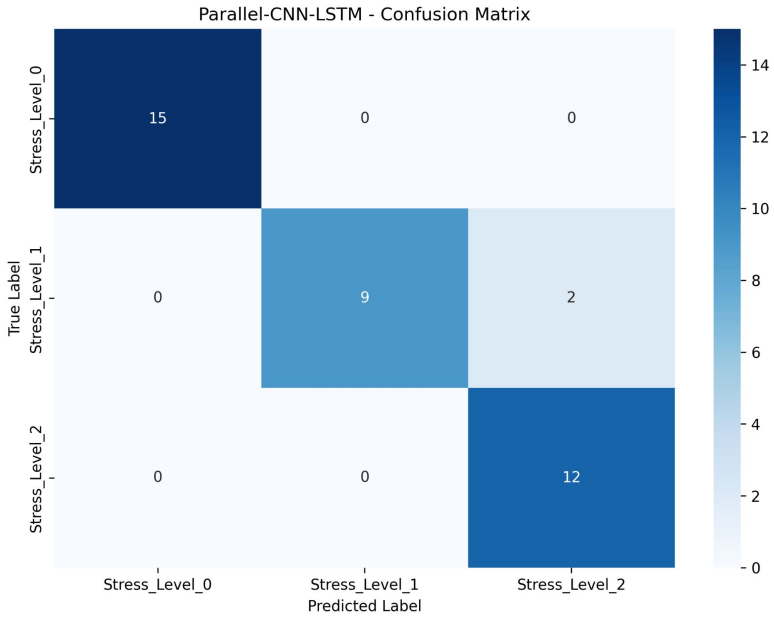
Inception1D - LSTM



	precision	recall	f1-score	support
Stress_Level_0	1.00	1.00	1.00	15
Stress_Level_1	1.00	0.73	0.84	11
Stress_Level_2	0.80	1.00	0.89	12
accuracy			0.92	38
macro avg	0.93	0.91	0.91	38
weighted avg	0.94	0.92	0.92	38

Model Evaluation

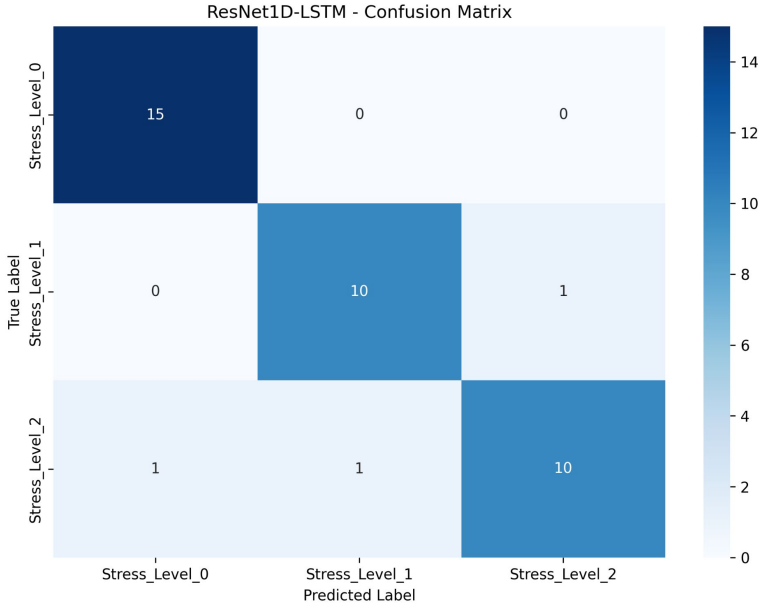
Parallel CNN - LSTM



	precision	recall	f1-score	support
Stress_Level_0	1.00	1.00	1.00	15
Stress_Level_1	1.00	0.82	0.90	11
Stress_Level_2	0.86	1.00	0.92	12
accuracy			0.95	38
macro avg	0.95	0.94	0.94	38
weighted avg	0.95	0.95	0.95	38

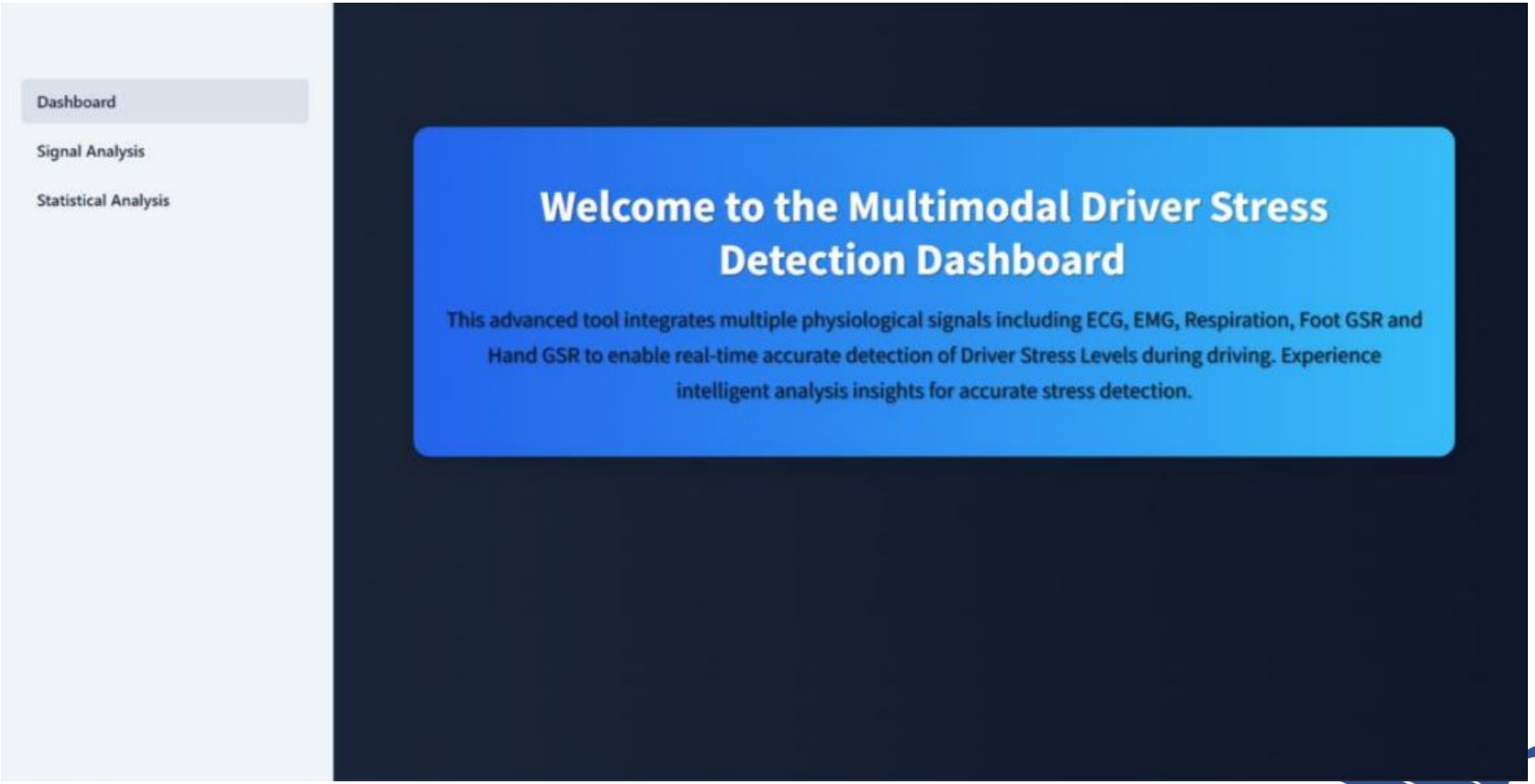
Model Evaluation

Resnet1D - LSTM



	precision	recall	f1-score	support
Stress_Level_0	0.94	1.00	0.97	15
Stress_Level_1	0.91	0.91	0.91	11
Stress_Level_2	0.91	0.83	0.87	12
accuracy			0.92	38
macro avg	0.92	0.91	0.92	38
weighted avg	0.92	0.92	0.92	38

Website


Dashboard
Signal Analysis
Statistical Analysis
Welcome to the Multimodal Driver Stress Detection Dashboard
This advanced tool integrates multiple physiological signals including ECG, EMG, Respiration, Foot GSR and Hand GSR to enable real-time accurate detection of Driver Stress Levels during driving. Experience intelligent analysis insights for accurate stress detection.

Website

Dashboard

Signal Analysis

Statistical Analysis

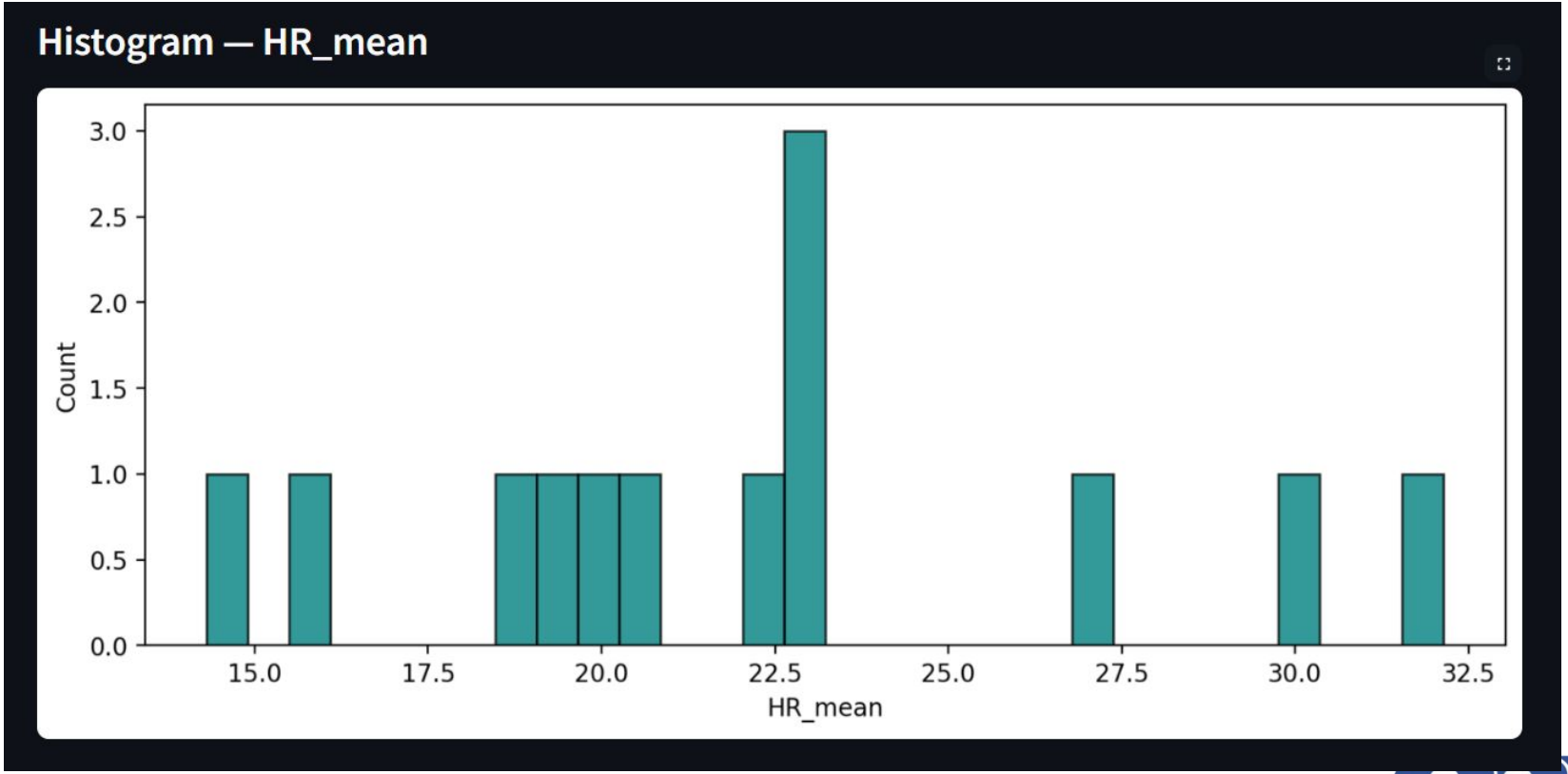
Statistical Analysis of Extracted Dataset

Dataset Shape: 259 samples, 30 columns

Descriptive Statistics

	count	mean	std	min	25%	50%	75%	max
HR_mean	259	36.1137	17.6714	6.1394	21.415	33.4313	44.8204	79.4463
SDNN	259	2017.4174	1883.1529	73.8139	802.9439	1368.5286	2920.4086	15196.2456
RMSSD	259	2841.7549	2659.782	98.2102	1103.671	1885.7594	4051.8694	20700.4326
pNN50	259	0.9126	0.1114	0.4961	0.9167	0.9451	0.9718	1
LF_HF	259	3.7902	11.4837	0	1.3909	1.634	2.318	119.8917
RR_mean	259	2.1291	1.1642	0.7552	1.3387	1.7947	2.8018	9.773
RR_std	259	2.0174	1.8832	0.0738	0.8029	1.3685	2.9204	15.1962
RR_min	259	0.628	0.0212	0.6	0.62	0.63	0.63	0.77
RR_max	259	12.2458	10.0577	1.32	5.26	9.55	16.26	60.07
EMG_mean	259	0.5637	0.605	0	0.0624	0.29	0.9377	3.2141

Website


Histogram — HR_mean
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
15.0
17.5
20.0
22.5
25.0
27.5
30.0
32.5
HR_mean

Inferences

- Multimodal biosignals (ECG, EMG, GSR, respiration) outperform unimodal approaches, especially in noisy real-world driving.
- Classical ML models offer interpretability but struggle with long-term temporal patterns, limiting accuracy.
- Deep models (CNN-BiLSTM/GRU + Attention) achieve the best performance (up to ~97% accuracy) by capturing both local and long-term dependencies.
- BiLSTM gives highest accuracy but is computationally heavy; GRU offers near-similar performance with much better efficiency for real-time use.
- Most errors occur between medium vs. high stress, suggesting need for fuzzy/adaptive boundaries.
- The framework shows that deep multimodal fusion enables robust, scalable, real-time driver-stress monitoring for intelligent transportation systems.

Conclusion

This proposed multimodal driver stress detection framework highlights that carefully preprocessing biosignals and integrating them with deep temporal learning architectures leads to highly accurate, deployable systems for recognizing driver stress. CNN-BiLSTM-Attention sets a new benchmark of accuracy in offline evaluation, whereas CNN-GRU-Attention presents an efficient solution that can be practically adopted in real-world in-vehicle applications. Confirming strong generalization, the capability of this framework to deal with naturalistic, noisy physiological data supports the feasibility of AI-enabled invehicle driver stress monitoring for improving road safety and driver well-being in intelligent vehicles.

References

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THANK YOU!